CSE 4074 Computer Networks

Fall 2020 Midterm Exam

08.12.2020 Tuesday, Duration: 120 minutes (including upload process)

Q1	Q2	Q3	SUM
/50	/30	/20	/100

On my honor, I have neither given nor received any unauthorized assistance on this examination.

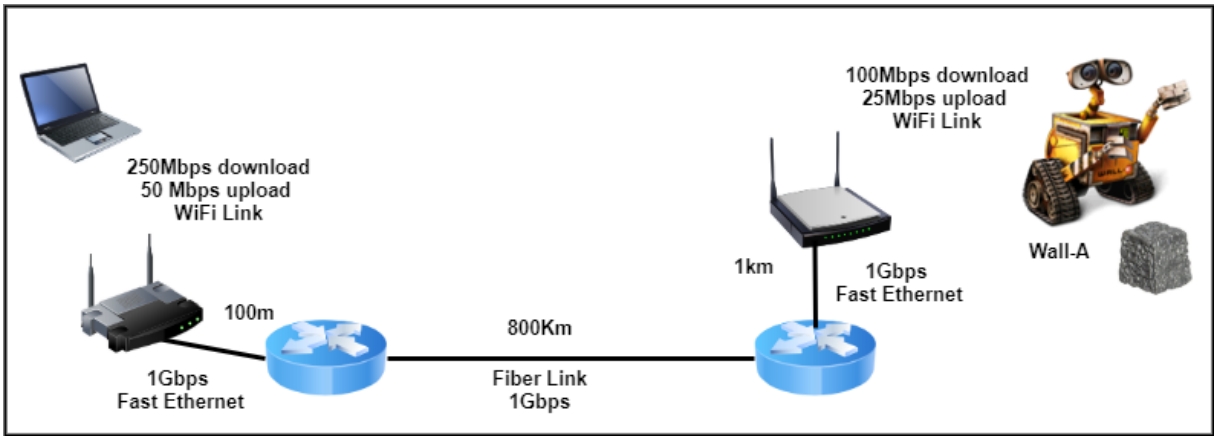
Name Surname:_____

Signature:_____

Solution and Submission Notes

- **Textbooks, slides, and notes are open.**
- **Internet usage is not allowed; however, computers and smartphones may be used for viewing course materials provided for this course.**
- **Write your name, surname and student ID on the top of each exam solution page.**
- **Show all your work.**
- **Write the following sentence on the top of the first page with your handwriting and sign it: "On my honor, I have neither given nor received any unauthorized assistance on this examination."**
- **If same mistakes repeat for two students, this can be considered as plagiarism.**
- **Either print this PDF file (6 pages) or use blank A4 pages for your solutions. Solve all the questions with your handwriting.**
- **Scan all the solution pages to a single PDF file named "firstname_lastname.pdf". Do not include raw photos, use an Android/IOS scanner application. All the writings should be very readable. Page layouts should be portrait, not landscape.**
- **Upload the PDF file to https://ues.marmara.edu.tr system before the deadline. Give at least 15-20 minutes for scanning and upload process. If you encounter any problem, you may upload via google classroom before the deadline.**

Q-1 (50 pts)



Imagine that you have a trash collecting robot Wall-A which can be controlled remotely (it is a primitive version of Wall-E). You can send remote commands to Wall-A such as “LEFT”, “RIGHT”, “FORWARD”, “BACKWARD”, “LIFT”, “RELEASE”. You have a laptop with a WiFi connection and it is connected to a router. Data rates and lengths of the links are given in the Figure. Speed of signals can be assumed to be 200.000 km/sec. You may assume that there is no network congestion and you may neglect queueing and processing delays. Give any assumptions that you make.

(a - 10 pts) Although it is not the best way to communicate, you decided to use HTTP for sending commands to Wall-A. You will implement a server process running in Wall-A, and you will be able to send commands from the browser. Describe how you can use HTTP to communicate with Wall-A. Which features of HTTP would you use? Give a sample request and a sample reply message.

(b - 10 pts) Assume that bit error rate is too low and packet loss probability can be neglected. Let’s say that you want Wall-A to receive the command at most in 10 ms. Can you achieve this using non-persistent HTTP connections? What about persistent HTTP connections? Describe your answer.

(c – 10 pts) Now assume that there could be some packet losses. You want to be sure about whether the packets are received by Wall-A or not in 10ms. However waiting for a timeout may be late, so you may want to take some actions in order to increase probability of receiving the command in time. Describe a protocol for achieving this using UDP in the transport layer. (You may design your own protocol)

(d – 6 pts) Now consider that Wall-A is employed in a large trash dump in Lagos, Nigeria and you want to control it from Istanbul. Again you want to send commands at most in 10 ms. A network company offers to upgrade your wireless technology and your ISP offers to double your link capacity. A satellite company offers to use its LEO satellite network. What can you do to send your commands in less than 10 ms delay?

(e – 8 pts) Let's say that Wall-A has a camera and provide real-time video streaming to the Laptop in the above figure. What is the maximum average throughput it can achieve? If connection quality of Wall-A continuously changes due to its movements, describe a method that makes use of available bandwidth (do not give name of any method, describe how to do in this case).

(f – 6 pts) Let's say that you do not apply end-to-end encryption while communicating with Wall-A. A bad guy wants to take control of Wall-A. Describe which method(s) he may apply to do this.

Q-2. (30 pts) You own a company named trashcollector. The following table gives DNS records for trashcollector.

Name	Value	Type	TTL (sec)
trashcollector.com	192.143.34.145	A	86400
wall-a.trashcollector.com	192.143.34.132	A	86400
ns.trashcollector.com	191.143.34.2	A	86400
server1.trashcollector.com	192.143.34.135	A	86400
trashcollector.com	ns.trashcollector.com	NS	86400

(a – 3 pts) Which of the above resource records are stored in the TLD DNS server?

(b – 4 pts) server1.trashcollector.com is your web server. You want your customers to access web via www.trashcollector.com as well. Which resource record(s) would you add to your authoritative DNS server?

(c – 5 pts) Can you run a mail server on the same host as the web server (server1.trashcollector.com)? Justify your answer. If your answer is no, you may give any name and IP address for the mail server. You want to use mail addresses such as username@trashcollector.com. Which resource record(s) you would add to your authoritative DNS server?

(d – 6 pts) Suppose that you plan to change the IP address of your web server after 1 week. What action(s) you may take in order to minimize the interruptions that your customers experience? Describe why your solution works.

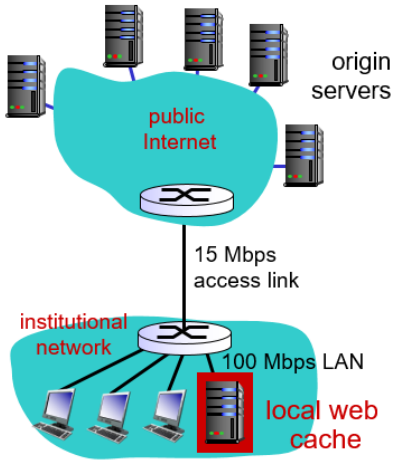
(e – 6 pts) A study on the traffic at root DNS name servers shows that 95% of the queries they receive are for non-existent top level domains, e.g., typos such as marmara.edu.ty (.ty domain does not exist) instead of marmara.edu.tr. On the other hand, a study of a representative cross-section of local DNS servers shows that only 0.1% of the queries they receive are for non-existent top level domains. How can you explain this discrepancy?

(f – 6 pts) A modification is proposed for DNS so that local name servers will also cache negative results, i.e., names or domains that could not be resolved, in order to reduce the query traffic to root servers. Will this modification be more effective if the erroneous queries for non-existent domains are generated mostly by users typing wrong URLs into their browsers by mistake, or if they are generated by malicious programs repeatedly trying to resolve a non-existent domain name? Describe your answer.

Q-3. (20 pts) Consider the institutional network shown in the figure below. Suppose that average object size is 540,000 bits and that the computers in the institution send out 25 HTTP requests per second. Also suppose that the amount of time it takes from when the router in the Internet side of the access link forwards an HTTP request until it receives the response is 2 seconds on average. You may model the total average response time as the sum of the average access delay (delay from Internet router to institution router) and the average Internet delay.
The average access delay is related to the traffic intensity as given in the folowing table.

Traffic Intensity	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
Average access delay (msec)	26	33	41	52	64	80	100	137	250	1000

Remember that traffic intensity is calculated as follows: Traffic intensity = aL/R, where a is the arrival rate, L is the packet size and R is the transmission rate.



(a – 10 pts) Find the total average response time when the cache is not active.

(b – 10 pts) Now, suppose that a cache is installed in the institutional LAN, and suppose the hit rate is 1/3. Find the average response time.